



Skeletal transformation

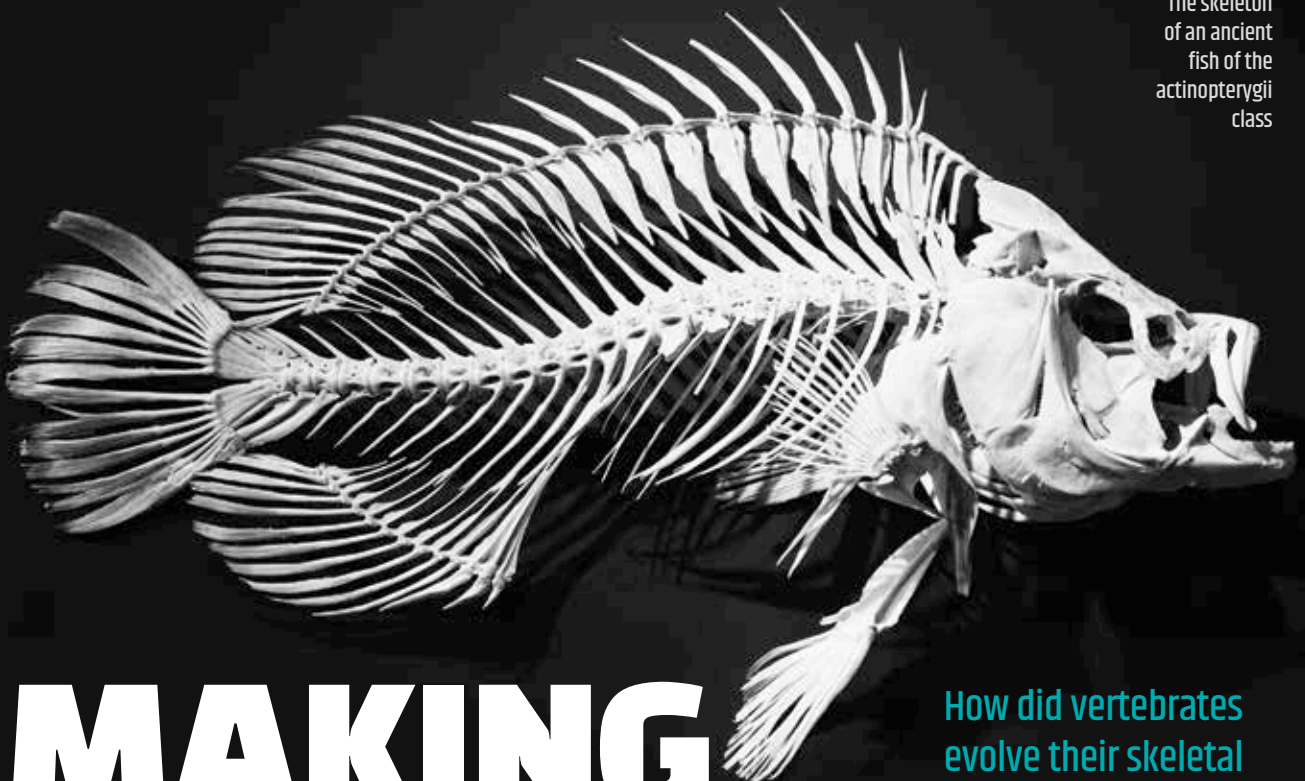
A look at all the various ways that modern life is causing changes in the skeletons of humans. <https://digit.in/apr20-59>



The gift of the hyoid

Sapiens and Neanderthals have a hyoid bone, the only structure that allows for articulate vocalisation. <https://digit.in/apr20-60>

The skeleton of an ancient fish of the actinopterygii class



MAKING BONES

How did vertebrates evolve their skeletal systems? There are no bone-afied answers.

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he skeletal system is what provides structure and support to the animal bodies.

Insects and crustaceans have an exoskeleton, which is a protective shell outside the body. Vertebrates have an endoskeleton, which is inside of the body. The tortoise has both an exoskeleton as well as an endoskeleton. There are like four types of hardened tissues that form the skeleton in creatures – enamel, teeth, bones and cartilage. The word skeleton originates from skeletós, an ancient Greek word for dried up or withered.

For most of history, academic research into this topic has been centered around the study of fossilised animals. The problem is that the fossil record

is incomplete, and it is difficult to ascertain which specific type of skeletal tissue is preserved in the fossil. The most primitive organisms are less fully preserved, exacerbating the problem of understanding the origins and evolution of the skeletal system. Ernst Haeckel, one of the contemporaries of Charles Darwin, used embryology to understand the evolution of animals. However, Haeckel and Darwin disagreed on one of the aspects of evolution. While Haeckel believed that evolution was a progression from simple to more complex forms, Darwin did not believe that evolution was progressive, and that animals merely adapted to their surroundings, the complications being kind of like a side effect, inconsequential and irrelevant. Haeckel

additionally believed in Lamarckism, another alternative version of evolution where animals pass on acquired traits to their progeny. This means that if you physically stretch apart a human's neck, or bind the feet, their progeny will have elongated necks or tinier feet. Haeckel's belief in both Darwinism and Lamarckism resulted in what is known as the biogenetic law, summarised by the phrase "ontogeny recapitulates phylogeny". The idea is that the development of the embryo mirrors the evolution of the creature, with each point in the embryological development corresponding to the adult form of an ancestor. While the theory was influential for decades, it has now been discarded by scientists. The most compelling evidence refuting the